

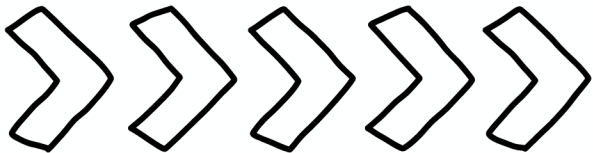
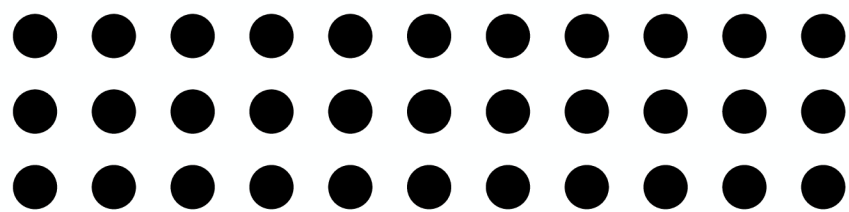
May 18, 2026



# WorkPulse

## Employee Retention Analytics Platform

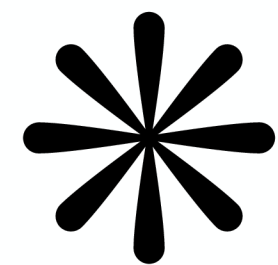
An-Najah National University 



# \*Problem Statement

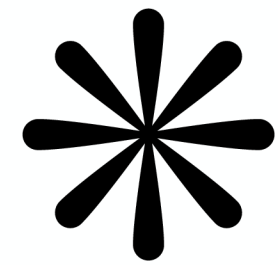
According to Gallup reports and recent workplace studies, employee engagement and retention remain major challenges for modern organizations. Only **20% of employees worldwide are actively engaged at work**, while nearly **41% experience daily workplace stress**. In addition, more than **50% of employees report symptoms of burnout**, leading to reduced productivity, lower job satisfaction, and increased employee attrition. These issues significantly increase hiring and training costs and negatively impact organizational stability and overall business performance.





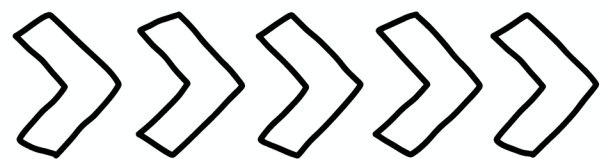
# Project Idea

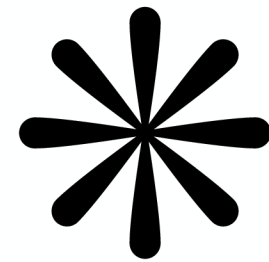
This project aims to develop an intelligent Python-based HR analytics platform that analyzes employee data to detect attrition risks, identify burnout patterns, and segment employees into meaningful groups. The system also provides smart HR recommendations and interactive dashboards to help organizations improve employee retention, enhance workforce performance, and support data-driven decision-making.



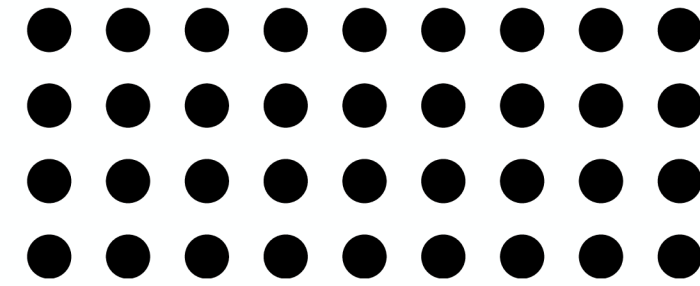
# Project Goals

- Analyze HR data using data mining techniques
- Segment employees into meaningful workforce groups
- Detect attrition, burnout, and disengagement risks
- Identify unusual employee behavior patterns
- Generate intelligent HR recommendations
- Support data-driven HR decision making
- Improve employee retention and workforce performance





# The Process



1

**Dataset Upload:** CSV upload → Validation → Dataset Preview

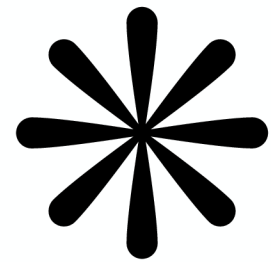
2

**Data Cleaning:** Remove duplicates → Handle missing values → Encode categorical features → Scale numerical features → Save multiple dataset versions

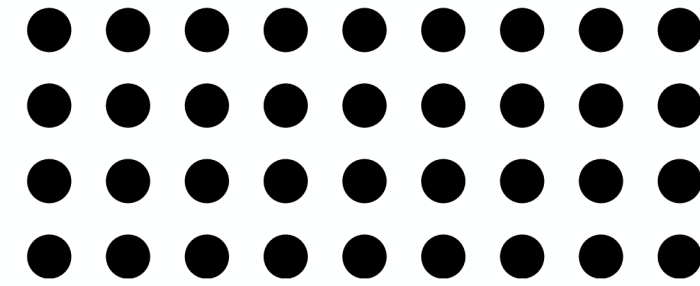
3

**Auto Column Detection:** Detect important HR columns automatically → Support different company datasets





# The Process



4

**Segmentation Pipeline:** Feature Selection → StandardScaler → PCA (SVD) → KMeans

5

**Evaluation Metrics:**

Best K

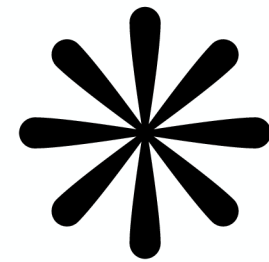
Silhouette Score

Davies-Bouldin Index

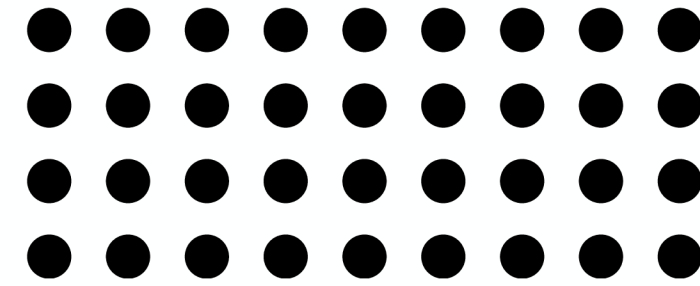
Calinski-Harabasz Score

6

**Risk Detection:** uses the segmented dataset as input → calculates attrition and burnout scores → uses Isolation Forest → classifies employees into risk categories → adds clear danger explanation



# The Process

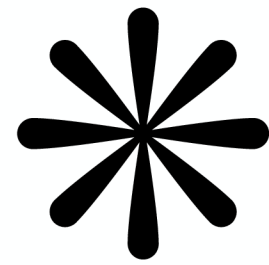


7

**Recommender System:** Builds an employee profiles using HR features → normalizes employee features → uses cosine similarity → compares with healthy peers → Generates personalized HR actions and retention strategy

8

**User Interface and Dashboard:** Each team member designed and implemented the UI/dashboard interface for their own module independently, while maintaining integration consistency across the platform.

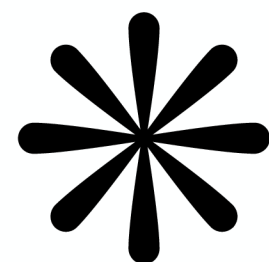


# Conclusion

- The project demonstrates how data mining and AI-driven analytics can help HR departments proactively detect attrition and burnout risks before they escalate.
- By combining predictive scoring, employee segmentation, anomaly detection, and a content-based recommendation engine, the platform provides personalized and actionable HR insights.
- Overall, the platform transforms raw employee data into meaningful intelligence that enhances workforce stability and organizational performance.







**Thank  
You**

